

Exercise Sheet: Crash Course in Calculus & Linear Algebra

Math Recap

This exercise sheet reviews the core concepts from single- and multivariable calculus and linear algebra that are needed to understand the Laplace operator conceptually. The exercises build on each other: from scalar differentiation rules through partial derivatives to eigenvalue problems, which directly motivate the structure of the Laplace operator.

Part I Rules Overview

Differentiation Rules (one-dimensional)

Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and $c \in \mathbb{R}$.

Linearity: $(c f \pm g)' = c f' \pm g'$

Power rule: $(x^n)' = n x^{n-1}$

Product rule: $(f \cdot g)' = f'g + fg'$

Quotient rule: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g \neq 0$

Chain rule: $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$

Standard derivatives:

$$(e^x)' = e^x \quad (\ln x)' = \frac{1}{x} \quad (\sin x)' = \cos x \quad (\cos x)' = -\sin x \quad (x^n)' = nx^{n-1}$$

Partial Derivatives & Gradient (multidimensional)

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be partially differentiable.

Partial derivative: $\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}$ (all other variables held constant)

Gradient: $\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$ (direction of steepest ascent)

Schwarz's theorem: $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$ (when mixed partials are continuous)

Integration by Parts

$$\int_a^b u(x) v'(x) dx = [u(x) v(x)]_a^b - \int_a^b u'(x) v(x) dx$$

Guideline for choosing u : Logarithm > Polynomial > Trigonometric > Exponential (LIATE).

Matrices, Vectors and Determinants

Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $\lambda \in \mathbb{R}$.

Matrix multiplication: $(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$ (row $\times$ column; in general $AB \neq BA$)

Arithmetic rules:

$$\begin{aligned} (AB)C &= A(BC) \quad (\text{associativity}) & A(B+C) &= AB + AC \quad (\text{distributivity}) \\ (AB)^\top &= B^\top A^\top & (A^{-1})^{-1} &= A \end{aligned}$$

Determinant ($A \in \mathbb{R}^{n \times n}$):

$$\det(2 \times 2): \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$\det(3 \times 3)$ (cofactor expansion along row 1): $\det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$
where M_{ij} denotes the (i, j) -minor determinant.

Properties of the determinant:

$$\det(A^\top) = \det(A) \quad \det(AB) = \det(A) \det(B) \quad \det(\lambda A) = \lambda^n \det(A)$$

Row swap $\Rightarrow$ sign changes Zero row/column $\Rightarrow \det = 0$

$$\det(A) \neq 0 \iff A \text{ is invertible}$$

Eigenvalues and Eigenvectors

$\lambda \in \mathbb{R}$ is called an **eigenvalue** of $A \in \mathbb{R}^{n \times n}$ if there exists a vector $\mathbf{v} \neq \mathbf{0}$ such that:

$$A \mathbf{v} = \lambda \mathbf{v}$$

Computation:

$$\det(A - \lambda I) = 0 \quad (\text{characteristic polynomial}) \Rightarrow \lambda_1, \dots, \lambda_n$$

Eigenvectors: for each λ_i , solve $(A - \lambda_i I) \mathbf{v} = \mathbf{0}$.

Key properties:

- Symmetric matrices ($A = A^\top$) have only real eigenvalues.
- Eigenvectors of a symmetric matrix for distinct eigenvalues are orthogonal.
- $\det(A) = \prod_i \lambda_i$ and $\operatorname{tr}(A) = \sum_i \lambda_i$.

Part II Exercises

Block 1: Differentiation Rules & Polynomials

Exercises 1–4.

Exercise 1 Power Rule & Linearity

Compute the derivative of the following function and simplify your result:

$$f(x) = 3x^4 - 5x^2 + 7x - 2$$

Hint: Apply the power rule $(x^n)' = nx^{n-1}$ term by term.

Exercise 2 Product Rule

Compute $g'(x)$ for

$$g(x) = x^2 \cdot e^x$$

and factor the result. *Hint: Product rule: $(f \cdot h)' = f'h + fh'$.*

Exercise 3 Chain Rule

Compute the derivative of:

$$h(x) = \sin(x^3 + 1)$$

Explicitly identify the outer and the inner function.

Exercise 4 Combined: Quotient and Chain Rule

Compute:

$$k(x) = \frac{\ln(x^2 + 1)}{x}, \quad x \neq 0$$

Hint: Combine the quotient rule with the chain rule for $\ln(\cdot)$.

Block 2: Partial Derivatives & Gradient

Exercises 5–7.

Exercise 5 First-Order Partial Derivatives

Find all first-order partial derivatives of:

$$f(x, y) = x^2y + y^3 - 3xy.$$

Exercise 6 Second-Order Partial Derivatives (Hessian)

Compute all second-order partial derivatives of

$$f(x, y) = x^3 + x^2 y^2$$

and assemble the Hessian matrix:

$$H_f(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

Verify Schwarz's theorem.

Exercise 7 Gradient in $\mathbb{R}^3$

Compute the gradient of

$$f(x, y, z) = x^2 y + y z^2$$

at the point $\mathbf{p} = (1, 2, 1)^\top$. Interpret the result geometrically: what does the gradient describe?

Block 3: Integration by Parts**Exercises 8–9.****Exercise 8**

Evaluate the indefinite integral:

$$\int x e^x \, dx$$

State your choice of u and v' explicitly.

Exercise 9

Evaluate:

$$\int x \sin(x) \, dx$$

Hint: This integral appears frequently in Fourier analysis.

Block 4: Matrices, Vectors and Determinants**Exercises 10–13.****Exercise 10 Matrix Multiplication**

Compute the product $A \cdot B$ for

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then check whether $AB = BA$ holds.

Exercise 11 Determinants

(a) Compute the determinant:

$$\det(A) = \det \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

(b) Using cofactor expansion along the first row, compute:

$$\det(B) = \det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$$

(c) What does $\det(A) \neq 0$ imply for the linear system $A\mathbf{x} = \mathbf{b}$?

Exercise 12 Properties of the Determinant

Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Verify the following properties by direct computation:

- (a) $\det(A^\top) = \det(A)$
- (b) $\det(AB) = \det(A) \cdot \det(B)$
- (c) $\det(2A) = 2^2 \cdot \det(A)$

Exercise 13 Cross Product

Compute $\mathbf{u} \times \mathbf{v}$ for

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Hint: The cross product can be computed as the determinant of a 3×3 matrix with unit vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in the first row. Interpret the result geometrically.

Block 5: Eigenvalues (real, two-dimensional)**Exercises 14–15.****Exercise 14 Eigenvalues and Eigenvectors**

Consider the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (a) Find the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
- (b) Compute the eigenvalues λ_1, λ_2 .
- (c) Find the corresponding eigenvectors.
- (d) Check whether the eigenvectors are orthogonal. What is the geometric reason?

Exercise 15 Rotation Matrix – Limits of Real Eigenvalues

Consider the 90 rotation matrix:

$$B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (a) Compute $\det(B - \lambda I)$ and solve the characteristic polynomial.
- (b) What do you observe? What is the geometric interpretation?
- (c) **Connection to calculus:** The equation $f''(x) = \lambda f(x)$ is the direct analogue of the eigenvalue problem $A\mathbf{v} = \lambda\mathbf{v}$, but for the operator $\frac{d^2}{dx^2}$. Which functions f satisfy this equation for $\lambda < 0$? (Justify your answer; no formal proof required.)

Outlook: The Laplace Operator

All concepts from the preceding exercises converge in the **Laplace operator** Δ . It is defined as the sum of all second-order partial derivatives of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$:

$$\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \cdots + \frac{\partial^2 f}{\partial x_n^2}$$

In $\mathbb{R}^3$ explicitly:

$$\Delta f(x, y, z) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Conceptual meaning. The gradient ∇f tells you *where* a function is increasing (direction of steepest ascent). The Laplace operator $\Delta f = \operatorname{div}(\nabla f)$ measures how much the value of a function at a point deviates from the *local average* of its neighbourhood – it quantifies the mean curvature in space:

- $\Delta f(\mathbf{x}) > 0$: the point lies below its local average (local minimum / “valley”)
- $\Delta f(\mathbf{x}) < 0$: the point lies above its local average (local maximum / “peak”)
- $\Delta f(\mathbf{x}) = 0$: *harmonic function* – the point is exactly the local mean

Connection to eigenvalues (Exercise 15b). The equation

$$\Delta f = \lambda f$$

is the eigenvalue problem of the Laplace operator. Functions satisfying this equation for $\lambda = -\omega^2 < 0$ are sine and cosine functions – the building blocks of Fourier analysis. This is the starting point for the heat equation, the wave equation, and large parts of numerical mathematics and signal processing.
